Supplementary File S6: Integrated Simulator Run

TABLE I
GROUPED CHECK OUTCOMES FROM THE INTEGRATED RUN.

Pipeline stages	Pass	Warn	Fail	Step purpose
P0–P2	11	0	0	Bootstrap, model reference, and provisioning
P3, P4, P4m, P4b–P4f	30	0	0	Acquisition, operation, governance, and records
P6–P7	7	0	0	Seeded objective and multimodal acquisition checks
P8, P8b	16	0	0	Second-scene check and bounded-backstop drill
P9	3	0	0	Platform subsystem checks
P10	9	1	0	BOPET line provisioning and scripted task path
Total	76	1	0	17 phase gates passed

TABLE II
ENDPOINT CHECKS FOR THE IN-RUN `film-line` CONFIGURATION.

Protocol	Samples	Rejected probes	Legal-write check
Modbus TCP	12	3/3	One accepted
OPC UA	12	3/3	One accepted
MQTT	5	3/3	One accepted
Modbus RTU	11	–	Acquisition only
HTTP	11	–	Acquisition only
Total	51	9/9	Three accepted checks

TABLE III
PROTOCOL-LEVEL MEASUREMENTS RECORDED IN THE INTEGRATED RUN.

Protocol	Samples	Rejected probes	Reported p_{50}/p_{95} (ms)
Modbus TCP	9	6/6	47.673/48.348
OPC UA	11	6/6	21.494/24.175
MQTT	6	6/6	23.015/94.108
HTTP	13	6/6	34.414/47.784
Modbus RTU	15	–	–

This supplement covers only the integrated run reported in Section V. The accompanying ZIP contains selected machine-readable records, source tables, and figure assets; it is not a runtime image.

I. Run Conditions and Phase Checks

The run used a deterministic cast-film simulator and tool adapter; no language-model loop or physical equipment was used. The platform and simulator were already running, and a shared data directory was reused. The run identifier was `20261031034253-1dbo`. The command was `node bench/pipeline.mjs --profile integrated --seed 42 --cl-seeds 3`. The recorded runtime was Node.js 24.19.0 on Windows x64; the recorded source commit was `60fa983e`. All reported measurements refer to this recorded source snapshot; they are not a fresh validation of the repository's current HEAD.

All 17 phase gates reported PASS. The underlying checks comprised 76 PASS, one WARN, and zero FAIL outcomes. The warning was the BOPET semantic-card check in P10; it did not fail the phase gate. Table I groups the checks in execution order.

II. Acquisition, Write Probes, and Latency

The primary cast-film scene used five simulated devices across Modbus TCP, OPC UA, MQTT, HTTP, and Modbus RTU. Five acquisition nodes returned 54 samples. Modbus RTU was an acquisition endpoint without a write-control node. Each of the other four protocol paths received six negative probes; all 24 were rejected by the configured write-admission checks. Each of the four writable paths logged one accepted legal write. These are positive-path checks, not a complete denominator for estimating false-block rates. The second scene configuration (`film-line`) was checked within the same run across five endpoints (Modbus TCP, OPC UA, MQTT, Modbus RTU, and HTTP), returning 12, 12, 5, 11, and 11 samples, respectively (51 total). Its three writable paths each logged one accepted legal write and rejected three negative probes (9/9 total); no false block or code change was recorded. This is an in-run configuration check, not an independent deployment or repetition, and the observations do not estimate a general false-block rate.

The latency entries are per-protocol summaries stored by the run. The log describes approximately six writes but does not preserve exact per-protocol latency denominators or event-level timing observations. The run summary's 31.649 ms field is the unweighted arithmetic mean of the four per-protocol p_{50} values, not a pooled median. Latency values are therefore descriptive and do not support adapter rankings or population-level tail estimates.

III. Seeded Objective and BOPET Task

The three fixed-seed objective cases used the simulator's offline reference $J^* = 89.894$. The simulator score is $J = 55J_{\text{thick}} + 25J_{\text{quality}} + 8J_{\text{energy}} + 7J_{\text{throughput}}$. The thickness term scores deviation from $50 \pm 2\,\mu\text{m}$ with soft degradation outside the band; the quality term reflects defect rate, the energy term screw speed, and the throughput term line speed. The score is defined only for melt temperature in 195–225 °C and pressure no greater than 22 MPa. The runner reports J_{end} as the mean of the last two valid trajectory scores, using the sole valid score when a seed stops before a second score is available. Table IV also lists J_{last}, the final per-iteration score, to make the aggregation explicit. Seed 43's initial observation was inside the configured stopping region, so the

TABLE IV
FIXED-SEED SIMULATOR OBJECTIVE VALUES. J_0 IS THE INITIAL SCORE; J_{end} IS THE MEAN OF THE LAST TWO VALID TRAJECTORY SCORES, OR THE SOLE VALID SCORE IF THE SEED STOPS BEFORE A SECOND SCORE IS AVAILABLE; J_{last} IS THE FINAL PER-ITERATION SCORE RETAINED IN THE TRAJECTORY.

Seed	J_0	J_{end}	J_{last}	J^*	Ratio	Iter.	Writes
42	66.271	88.775	88.853	89.894	98.8%	2	2
43	86.542	86.542	86.542	89.894	96.3%	0	0
44	63.981	88.384	88.540	89.894	98.3%	2	2

run terminated with zero iterations and writes. This is a zero-update case; it provides no iterative-convergence evidence. The remaining seed cases used two writes each.

The BOPET trace was 28.05, 27.98, 27.98, and $24.84\,\mu\mathrm{m}$ across the initial state and three writes; the final reading was within the configured $25.0 \pm 0.7\,\mu\mathrm{m}$ band. The task board marked the task COMPLETED, but its associated parameter-change/optimization record remained open and unjudged. Treat this as an integration observation, not a completed or causally attributed optimization result.

IV. PROVENANCE AND INTERPRETATION

The source run lacked a contemporaneous SHA-256 manifest; ZIP hashes verify delivered files only, not runtime immutability.